FINAL PROJECT

Exploratory Data Analysis Report on Heart Attack Dataset

ALY6010: Probability Theory and Introductory Statistics

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Introduction

The study provides an exploratory analysis of a heart attack risk factor dataset. Patient attributes in the dataset consist of age along with cholesterol levels heart rate blood pressure smoking status diabetes history from 2019 to 2024. Analyzing this dataset aims to determine critical heart disease factors and examine patterns across various patient groups. Understanding these relationships enables us to discover insights which support early detection and prevention of heart disease.

This dataset comprises both numerical data and categorical information. Numerical characteristics consist of vital health measurements like Age and Cholesterol levels along with Blood Pressure and Heart Rate, while categorical data consists of patient Outcome (heart disease presence) and personal health history including Gender and Smoker status as well as Diabetes and Hypertension categories. Data cleaning required converting categorical variables to factor types while checking and identifying missing values and outliers as well as categorizing cholesterol levels into Normal, Borderline High, and High categories. The preprocessing steps prepare the dataset to support subsequent statistical analysis and visualization.

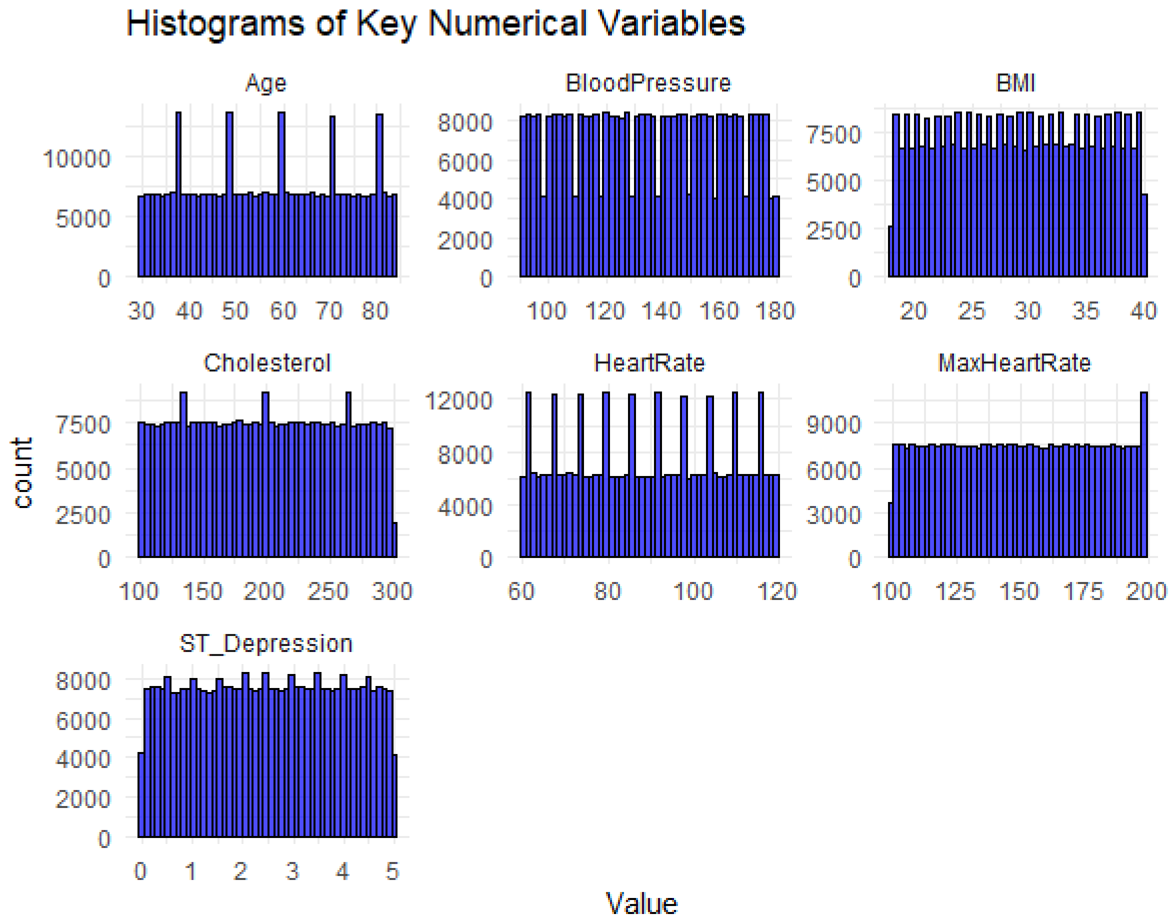
Data Analysis

Summary Statistics

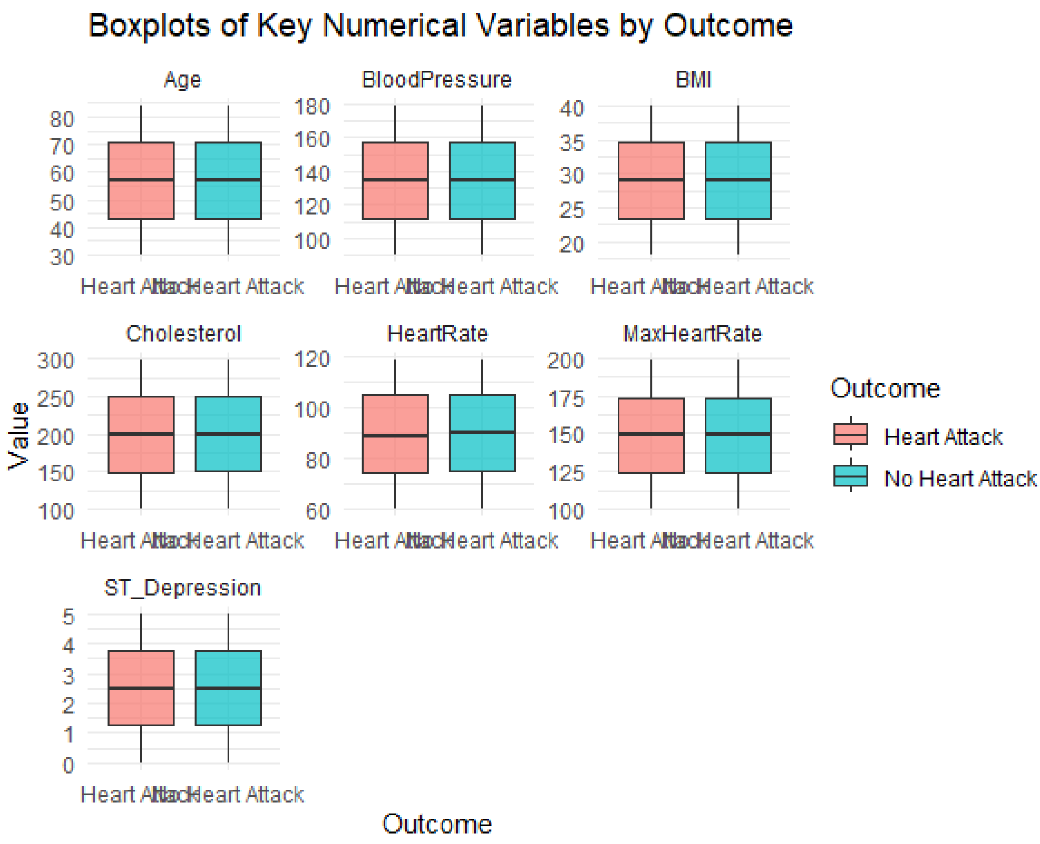
The broad range of cholesterol and blood pressure measurements demonstrates that patients face unique health challenges. The dataset clearly displays extreme values particularly in ST_Depression levels and cholesterol measurements that require further investigation. Analysis of the data reveals that heart disease manifests more often in older populations which validates previous research findings that illustrate the link between aging and cardiovascular risk factors.

Visualizations of Key Variables

The histograms I developed for numerical health variables enhanced my understanding of the distribution patterns of different health measurements. Cholesterol metrics display significant right skewness because a substantial number of patients have cholesterol values above 200 mg/dL which triggers health warning indicators. The range of blood pressure measurements across patients shows significant variation and includes some dangerously high readings.



Boxplots allowed comparison between heart disease patients and non-heart disease patients regarding the distribution of numerical variables. Individuals suffering from heart disease have elevated cholesterol and blood pressure readings compared to heart disease free individuals and they show varied BMI and MaxHeartRate patterns across groups. The differences in variable distributions prove that these factors serve a critical function in assessing heart disease risk.



Smoking and Cholesterol Levels

An in-depth subset analysis examined how smoking behaviors correlate with cholesterol measurements. The study separated cholesterol measurements into three categories and performed a cross-tabulation with smoking status(0=No, 1= yes). Smokers exhibit greater chances of having high cholesterol than non-smokers which is consistent with current medical studies demonstrating smoking increases heart disease risk. The creation of a bar chart demonstrated visually how smoking affects cholesterol levels which confirms smoking as a risk factor for heart disease through its impact on cholesterol.

	Smoker	Borderline High	High	Normal
1	0	37417	55993	93366
2	1	37260	55702	93236

Confidence Intervals and Hypothesis Testing

	Outcome	Age_CI_Lower	Age_CI_Upper	Cholesterol_CI_Lower	Cholesterol_CI_Upper	BloodPressure_CI_Lower	BloodPressure_CI_Upper
1	Heart Attack	56.88803	57.0322	199.0802	199.6048	134.4030	134.6393
2	No Heart Attack	56.91821	57.0621	199.3073	199.8307	134.3776	134.6131

HeartRate_CI_Lower	HeartRate_CI_Upper	BMI_CI_Lower	BMI_CI_Upper	MaxHeartRate_CI_Lower	MaxHeartRate_CI_Upper	ST_Depression_CI_Lower	ST_Depression_CI_Upper
89.39067	89.54834	28.98046	29.03813	149.3257	149.5879	2.495589	2.508700
89.44194	89.59912	28.97541	29.03294	149.3402	149.6017	2.494905	2.508005

To assess whether numerical variables significantly differ between heart attack and non-heart attack patients, I computed 95% confidence intervals for key metrics which include Age, Cholesterol, Blood Pressure, Heart Rate, BMI, MaxHeartRate, and ST_Depression.

From the table we can see, For ‘age, patients with heart disease have a 95% confidence interval of (56.89, 57.03), while those without heart disease have a 95% CI of (56.92, 57.06). As a result,‘age’ may not be a significant predictor of heart disease in this dataset. Cholesterol, Blood Pressure & Heart Rate have the same CI that may not be a strong differentiator. So I will try to seek their deeper relationship and show more useful information in my final report.

To determine whether the observed differences in cholesterol levels are statistically significant, we performed a two-sample t-test:

H0: There is no significant difference in mean cholesterol levels between heart disease and non-heart disease patients

H1: There is a significant difference in mean cholesterol levels between the two groups

Name	Type	Value
t_test_cholesterol	list [10] (S3: htest)	List of length 10
statistic	double [1]	-1.198042
parameter	double [1]	372968.5
p.value	double [1]	0.2309013
conf.int	double [2]	-0.597 0.144
estimate	double [2]	199 200
null.value	double [1]	0
stderr	double [1]	0.1890353
alternative	character [1]	'two.sided'
method	character [1]	'Welch Two Sample t-test'
data.name	character [1]	'df$Cholesterol[df$Outcome == "Heart Attack"] and df$Cholesterol[df$Outcome == " ...

From the table above we can see, p=0.231, which is greater than 0.05. This means there is not enough statistical evidence to reject the null hypothesis. So, cholesterol levels are not significantly different between heart attack and non-heart attack patients in this dataset.

Summary

The exploratory data analysis focused on determining important determinants of heart disease by evaluating patient data that included attributes such as cholesterol levels, blood pressure, heart rate, BMI, and ST Depression. We examined the differences in these variables between individuals with and without heart disease using descriptive statistics, visualizations, confidence interval analysis and hypothesis testing.

The study concluded that there was no significant statistical difference between cholesterol levels of the two groups examined. Analysis through the Welch two-sample t-test produced a p-value of 0.2309 which led us to maintain the null hypothesis. The significant overlap of confidence intervals for cholesterol between both groups demonstrates that cholesterol levels alone fail to strongly distinguish heart disease risk in this dataset.

Further examination of age-related data together with blood pressure readings and BMI and heart rate metrics showed that both groups exhibited only small variations. The single variable approach appears inadequate for accurate heart disease risk prediction while it becomes essential to consider the combined effects of multiple risk factors.

In conclusion, traditional heart disease risk factor such as cholesterol and blood pressure in this dataset cannot individually serve as strong predictor. To accurately determine heat attack risk we need an approach that examines multiple variables together. Research going forward needs to develop predictive models that incorporate various risk factors to achieve a thorough and precise understanding of heart disease prediction.